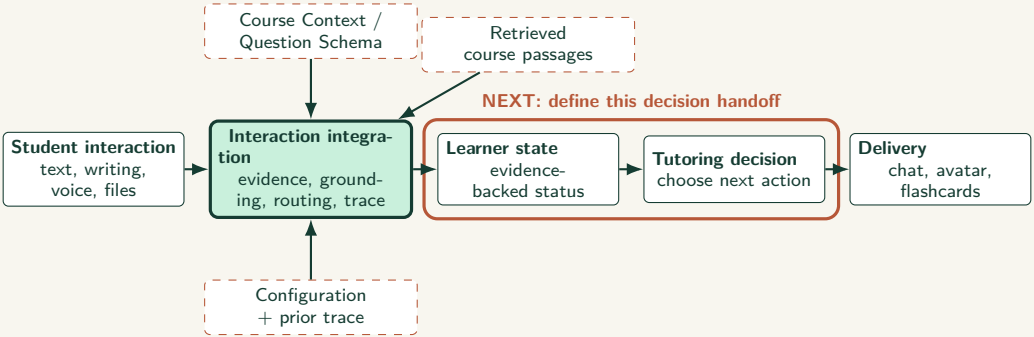
Interaction Integration Module Update

Inputs, functions, outputs, and downstream handoffs

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Where the interaction module fits



My module is the bridge: it converts student interaction into trusted evidence and moves that evidence, with its context, through the tutoring loop.

What comes in

Student-side inputs

- Text, equations, handwriting, voice, images, or files
- Recognized content and possible alternatives
- Student corrections, explanations, and uncertainty
- Session, question, and turn identity

System-side inputs

- Course Context and Question Schema
- Retrieved passages and page citations
- Previous learner state and prior tutoring action
- Tutoring rules and delivery options

Key rule: preserve where every item came from so later decisions can be explained and checked.

What the module does

1. Understand the submission

1. Identify the input type
2. Run the appropriate recognition tool
3. Ask the student about meaning-changing ambiguity
4. Capture work, answer, explanation, and uncertainty

2. Prepare and route evidence

1. Link the submission to its question and prior help
2. Request supporting course material
3. Validate the evidence record
4. Send it to the learner-state component
5. Record the resulting action and response

Boundary: this module prepares and routes evidence; the downstream components interpret it and choose the tutoring action.

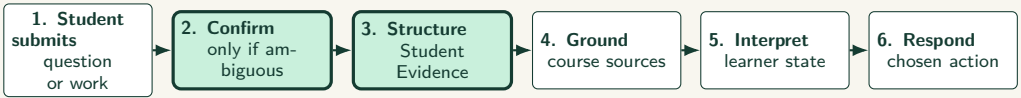
What goes out — and where it goes

Output	Receiver	Purpose
Retrieval request	Course Knowledge/RAG	Find relevant course passages and citations.
Student Evidence	Learner State	Determine what the student has demonstrated or still needs.
Grounded turn context	Tutoring Decision	Choose a question, hint, explanation, retry, or reassessment.
Session context	Tutoring Delivery	Present the chosen action through chat, avatar, or another channel.
Interaction trace	Persistent State	Preserve the sequence for continuity and later review.

Common contract

Each handoff includes identity, source, validation status, and the context version used to create it.

A tutoring turn from start to finish



The record grows

The system keeps the original submission, confirmed interpretation, sources, state change, action, and response as one linked history.

The loop continues

The student's next response creates new evidence linked to the earlier attempt and intervention. It does not overwrite either one.

What we need to define next

Primary focus

- What Learner State must contain
- How evidence changes that state
- What Tutoring Decision receives
- How it selects and explains an action

Immediate work

- Walk through one turn with all module owners
- Resolve missing or duplicated ownership
- Agree on minimum handoff fields and failure states
- Freeze interface version 0.1

Near-term result: a clear, reviewable contract from Student Evidence to Learner State to Tutoring Decision.